



EFFECTS OF VEGETATION COVER ON TEMPERATURE VARIATION WITHIN THE NIGERIA FOREST-SAVANNA ECOTONE

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Abstract

One of the most important benefits of natural vegetation in our environment apart, from it being the source of livelihood to man and other biological lives, is its role in modification of the local climate. This study investigated the role vegetation cover plays in regulating the micro-climate within the forest-savannah of Nigeria in relation to temperature, using geographic information system, climate and remotely sensed data. We employed descriptive statistics, linear regression analysis and trend analysis to unravel the dynamics of vegetation cover and temperature within the study area. The analysis of NDVI and Temperature from 2017 to 2022 reveals a stable profile, with both vegetation greenness (from 0.24 to 0.68) and temperature (from 24.25°C to 33.35°C) showing slight seasonal or inter-annual variations. The linear regression analysis further demonstrated that NDVI had a minimal and statistically insignificant influence on Temperature, as reflected in the low R^2 value (0.34%) and a p-value of 0.496 which is above the standard significance threshold (0.05). These findings suggest that temperature variations during this period were likely driven by other environmental factors not captured by NDVI alone. Future research may benefit from exploring additional variables to better understand the dynamics between vegetation health and climate in this ecosystem as this implicates regional food insecurity.

Keywords: NDVI, Temperature, Ecotone, Trends, Relationship and Remote Sensing

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Introduction

One of the most important benefits of natural vegetation in our environment apart, from it being the source of livelihood to man and other biological lives, is its role in environmental processes such as protection of soil against erosion, maintenance of carbon cycle and modification of the local climate (Osunmadewa & Wessollek, 2014). Historically, human environment had been subjected to changes by natural and anthropogenic factors which consisted mostly of transforming natural vegetation to settlements, farmlands and managed forest by man (Kastner *et al.*, 2011). These activities have caused notable consequences such as, loss of biodiversity, global warming, ozone layer depletion, amongst others (Olagunju, 2015). One of the factors that control changes in vegetation globally is climate (Kirilenko & Solomon, 1998). Climatic conditions such as rainfall and temperature also play important roles in vegetation changes and these changes which include but not limited to extreme dryness and wetness occur as a result of climate change (IPCC, 2012).

Climate change according to International Panel on Climate Change (2012) is as a result of extreme variability in weather phenomenon in terms of their frequency, spatial extent, intensity, duration, timing and their effects on human and other biodiversity. Irregular changes in precipitation and high temperature conditions were highlighted by Ezeaku *et al.*, (2014) as the major results of climate change. Wisam, (2015) stated that these two major factors play very important roles in changing the characteristics of the ecosystem. According to IPCC (2001), the impacts of climate change varies from region to region and is dependent on the nature, rate and severity of the climatic conditions in a particular region. They will also depend to an important degree on social, economic, governance and other forces that determine who and what are exposed to climate stresses, their sensitivities to stresses, and their capacities. Climate change is no longer a problem to be faced in

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the future, but a reality that is seriously affecting the earth already particularly challenging agricultural productivity and food security in developed and developing economies of the world (Piya, *et al.*, 2002).

The subject matter of climate change in Nigeria has been receiving keen interest among researchers in recent times, but most researches have been focused on the impacts of climate change on; weather parameters, agricultural activities, farmers' adaptation to the changes and food security (Sofoluwe, Tijani, & Baruwa, 2011; Obioha, 2009; Audu, Chukwu, Ogbonnaya, & Afuape, 2004; Ayoade, 2003). Different methodologies have been applied to the study of climate change. The methods ranges from use of basic climate-vegetation theories to modern use of remote sensing and GIS to delineate the dependence of vegetation on climate and the consequent effect of changes in climate on the same vegetation. Therefore, this study investigated the role vegetation cover plays in regulating the micro-climate within the forest-savannah of Nigeria in relation to temperature, using geographic information system, climate and remotely sensed data.

Methods

Study Area

The study area falls between latitude 6°0'0" N to latitude 7°09'0" N and longitude 6°36'0"E to longitude 7°42'0" E (Figure 1) which delineates the derived Savannah, which approximates the transitional zone between the southern rainforest and the tropical guinea savannah with a vegetation type that composed of a mixture of trees and grasses (Mayowa, Tadross, Abiodun & Omojola, 2010; Hoffmann & Jackson 2000; Morgan & Moss, 1965). The derived savannah also typifies the area referred to as wooded savannah (Adejuwon, 2006) and part of southern Guinea, a

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zone of moist peri-forest mixed with a savannah of anthropic degradation and patchy landscape (Hoffmann & Jackson, 2000; Bucini & Lambin, 2002). The land area is about 12,838.45km² (1,283,844.96 Hectares). The study area was delineated from an existing agroecological zone map adapted from the research paper of Keay, (1953) and Nwosu & Ugwuoti, (2009).

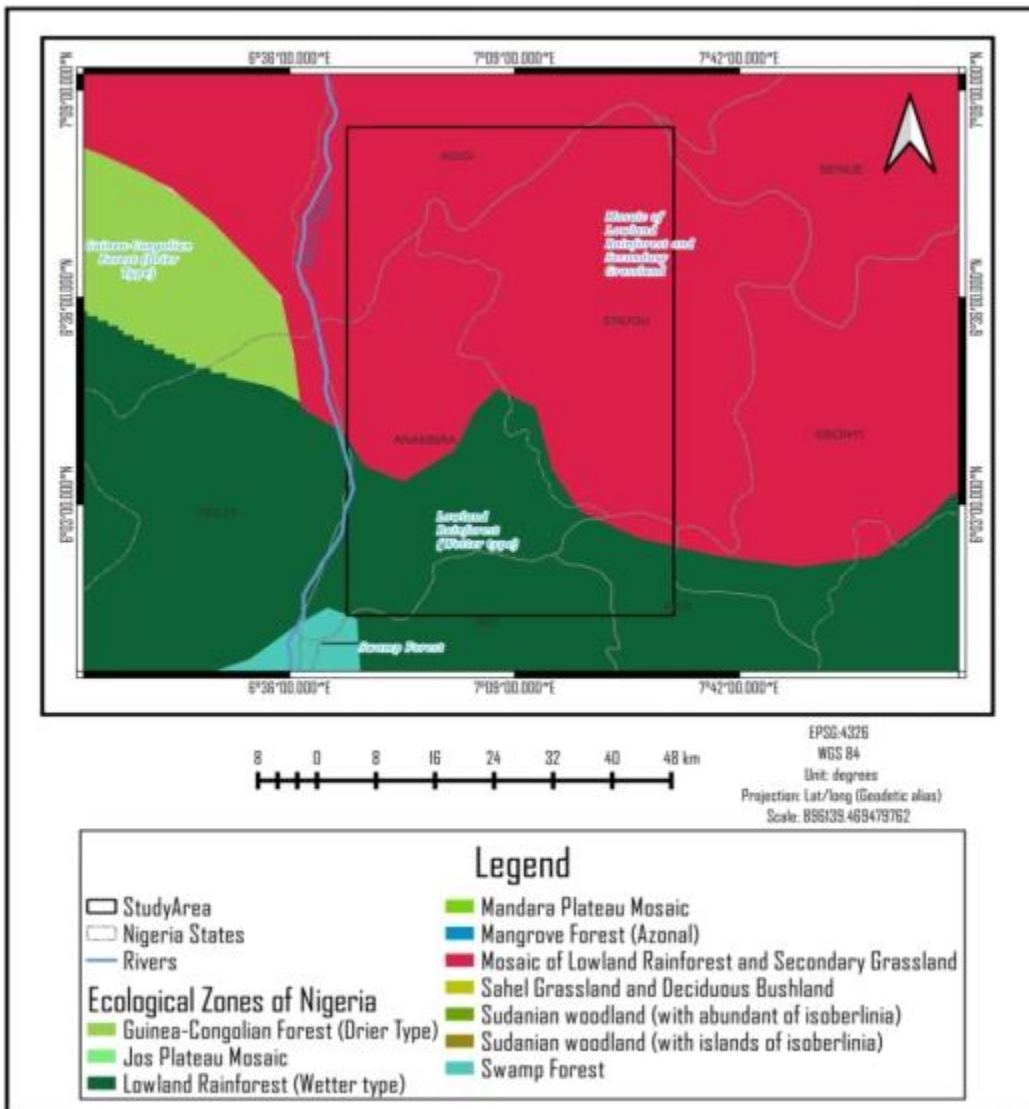


Figure 1: Study Area Map. The Black box delineates the study boundary

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Data source and processing

This research made use of secondary data. All dataset used in this study were geometrically referenced to the WGS 1984, UTM zone 32 projection system. The datasets include vegetation map of Nigeria, MODIS imageries and atmospheric data. The study period is from 2017-2022.

The NDVI is based on the fact that healthy vegetation has a low reflectance in the visible portion of the electromagnetic spectrum due to chlorophyll absorption and other pigments, and high reflectance in the Near Infrared (NIR) because of the internal reflectance by the mesophyll spongy tissue of a green leaf (Campbell, 1987). It also used to measure or estimate vegetation quantity and condition, thus it is a good indicator of climate change (Nduati, Mundia, & Ngigi, 2013). NDVI can be calculated using pixel's reflectance values of red (visible) band and the NIR band of a satellite sensor, and is represented by the following equation:

$$\text{NDVI} = \frac{\text{NIR} - \text{RED}}{\text{NIR} + \text{RED}} \dots\dots\dots \text{Equation 1}$$

NDVI values range from -1 to +1. Because of the high reflectance in the NIR portion of the electromagnetic spectrum, healthy vegetation is represented by high NDVI values between 0.5 and 1. Conversely, non-vegetated surfaces (such as water bodies) yield a negative NDVI value. Bare soil areas have NDVI values which are closest to 0 due to the high reflectance in both the visible and NIR portions of the electromagnetic spectrum (Lillesand & Kiefer, 1994).

To analyze the relationship between Temperature and NDVI from 2017 to 2022, descriptive statistics and Pearson correlation analysis were conducted. First, descriptive statistics were calculated to summarize the central tendency, dispersion, and range of each variable. These

International Journal of Climate Change and Environmental Sustainability statistics included the mean deviation, standard deviation, minimum, and maximum values, providing an overview of the dataset's characteristics over the selected timeframe. This step helped to establish a baseline understanding of the variability in Temperature and NDVI. The mean deviation ($\bar{X}$) is calculated as:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n |x_i - m(X)| \dots\dots\dots \text{Equation 2}$$

$m(X)$ = average value of the data set

n = number of data values

x_i = data values in the set

The standard deviation (s) measures the spread of the data around the mean, given by:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2} \dots\dots\dots \text{Equation 3}$$

The linear regression model was also adopted in analyzing the relationship between vegetation to climatic variables using NDVI and temperature as the index of measurement (Good & Caylor, 2011; Guo, *et al.*, 2014; Seddon, Macias-Fauria, Long, Benz, & Willis, 2016; Yang, *et al.*, 2016).

The linear regression model is mathematically presented as;

$$Y = \alpha + \beta x + \epsilon \dots\dots\dots \text{Equation 4}$$

Where Y represents Temperature, X represents NDVI, α is the intercept, β is the slope, and ϵ is the error term. The slope, β , indicates the change in Temperature for a one-unit change in NDVI. The

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model parameters α and β are estimated using the least-squares method, minimizing the sum of squared residuals:

$$\min \sum_{i=1}^n (Y_i - (\alpha + \beta X_i))^2 \dots\dots\dots \text{Equation 5}$$

The goodness-of-fit of the model is evaluated using the R-squared (R^2) value, which indicates the proportion of variance in Temperature explained by NDVI.

These analyses provide a comprehensive examination of the relationship between Temperature and NDVI, allowing us to understand central tendencies, correlation strength, and the predictive power of NDVI for Temperature in this dataset.

Results

The descriptive statistics for Temperature and NDVI over the period 2017–2022 provide insights into the central tendency, spread, and range of these variables within the dataset.

Table 1: Descriptive Statistics Results

	Temperature(°c)	NDVI
Mean	28.78	0.49
Std. Deviation	2.01	0.1
Minimum	24.25	0.24
Maximum	33.35	0.68

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For Temperature results in Table 1, the mean value is 28.78°C, indicating that average temperatures were relatively high over this five-year period. The standard deviation of 2.01°C suggests that temperature fluctuations were moderate, with most values falling within a relatively narrow range around the mean. The minimum recorded temperature was 24.25°C, while the maximum reached 33.35°C. This range of approximately 9°C shows that there were seasonal or inter-annual variations in temperature, but without extreme shifts.

NDVI (see table 1), an index used to measure vegetation greenness and density, has a mean of 0.49, suggesting moderate levels of vegetation cover over the time period. The standard deviation of 0.1 indicates low variability in NDVI values, implying relatively consistent vegetation density from 2017 to 2022. The minimum NDVI value of 0.24 and the maximum of 0.68, shows a moderate range, suggesting that vegetation productivity fluctuated but remained within certain bounds without extreme highs or lows. The narrow spread in NDVI values might also indicate either a stable ecosystem or limited influence of external factors that could lead to greater vegetation changes over time.

These values suggest that the period from 2017 to 2022 saw relatively stable conditions in terms of both temperature and vegetation greenness, with slight seasonal or environmental variations. The data implies that any significant shifts in vegetation activity or climate would likely be captured by more sensitive metrics, as these averages reflect fairly steady ecological conditions over the five-year period.

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To calculate a linear regression, there must be a linear relationship between the dependent and independent variables. In linear regression, a straight line is laid through the data; this only makes sense if there is linearity. The result of this is shown in figure 2.

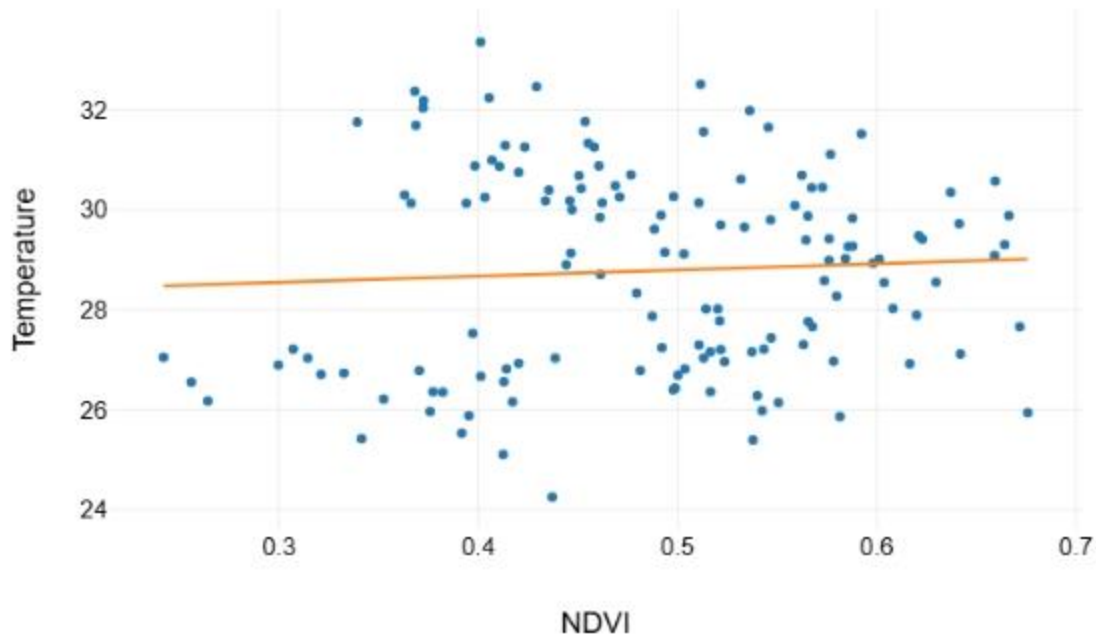
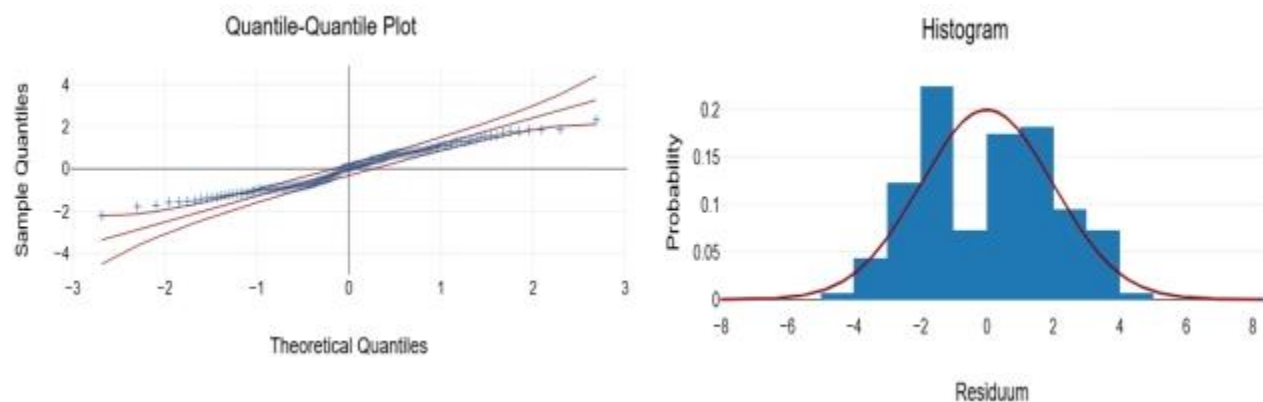


Figure 2: Regression graph of Vegetation against Temperature

We further tested for Normality of errors in our dataset and the results are presented in Table 2, Figure 3 ad 4.

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Table 2: Tests for normal distribution of Residuum

	Statistics	p
Kolmogorov-Smirnov	0.11	.064
Kolmogorov-Smirnov (Lilliefors Corr.)	0.11	<.001
Shapiro-Wilk	0.97	.003
Anderson-Darling	1.74	<.001



Figures 3 and 4: Residuum Histogram and Quantile-Quantile Plot

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Table 3: Durbin-Watson-Test

Autocorrelation	Statistics	p
0.79	0.41	<.001

The null hypothesis that the autocorrelation of the residual is zero is tested with a Durbin-Watson test. The calculated p-value of <.001 is smaller than 0.05, so that the null hypothesis is rejected and it can be assumed that there is autocorrelation of the residual.

In this study, conducted over the period from 2017 to 2022, we examined the relationship between NDVI (Normalized Difference Vegetation Index) and Temperature through a linear regression analysis. The primary aim was to assess whether NDVI, a widely used indicator of vegetation health, could explain variations in temperature. The results of the regression model indicated that NDVI explained only 0.34% of the variance in Temperature. This suggests that NDVI has a negligible effect on temperature variation, with other factors likely contributing more significantly to temperature changes. The very low R² value supports this interpretation, indicating that the model does not capture substantial variance in Temperature.

An ANOVA was conducted to determine whether the regression model significantly differed from zero. With an F-value of 0.47 and a p-value of 0.496, the results indicate that the effect of NDVI on Temperature is not statistically significant. In other words, the data do not support a meaningful

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relationship between NDVI and Temperature. The regression equation derived from the analysis was:

$$\text{Temperature} = 28.18 + 1.22 \cdot \text{NDVI}.$$

This equation suggests that if NDVI were zero, the predicted Temperature would be 28.18°C. Furthermore, a one-unit increase in NDVI is associated with a 1.22°C increase in Temperature. However, the p-value for the coefficient of NDVI was 0.496, well above the standard significance threshold of 0.05. This indicates that the coefficient is not significantly different from zero, suggesting that NDVI does not have a meaningful effect on Temperature in this sample. Additionally, the standardized regression coefficients show that NDVI has the largest influence on Temperature compared to other variables in the model. However, the lack of statistical significance in the p-value suggests that this observed influence is likely not meaningful or consistent across different samples. While the regression analysis yielded a mathematical relationship between NDVI and Temperature, the lack of statistical significance ($p = 0.496$) indicates that NDVI is not a reliable predictor of Temperature in this study. Over the period from 2017 to 2022, the data do not support the hypothesis that NDVI has a significant effect on temperature.

In addition to the descriptive and regression analysis, a trend analysis was conducted both for NDVI and temperature from 2017 to 2022 to observe potential patterns over time. The trend analysis plot (Figure 5 and 6) visually illustrates the fluctuations in NDVI and Temperature of the time period.

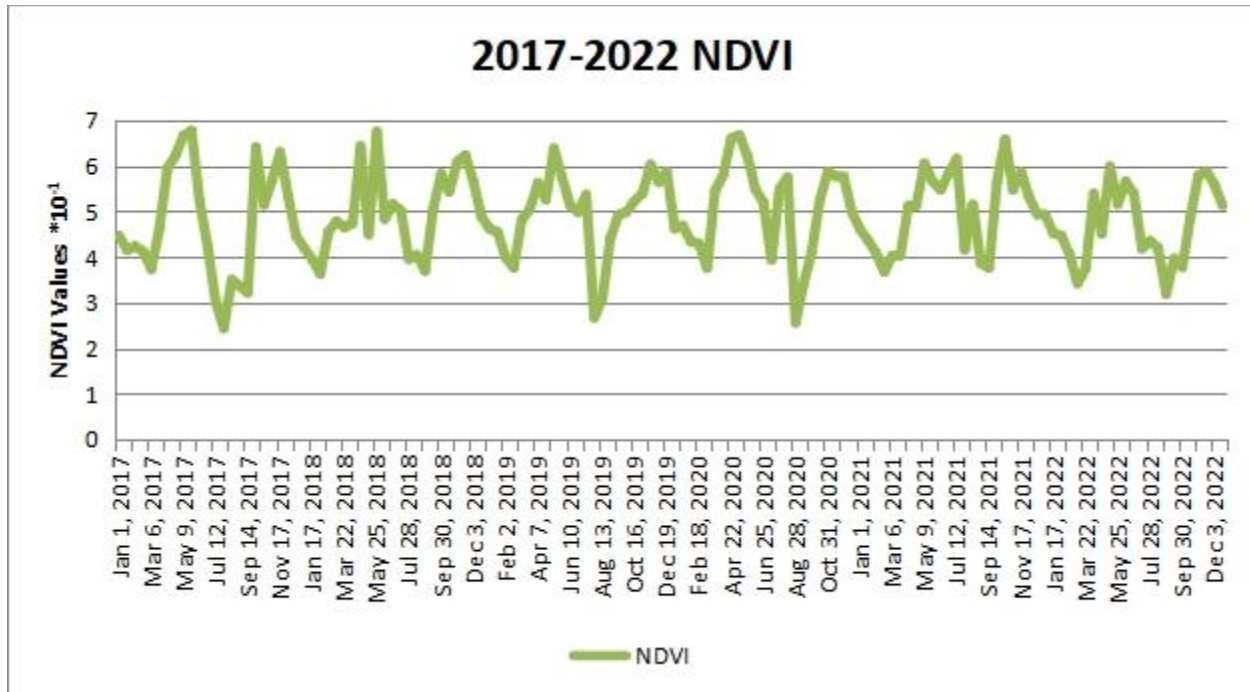
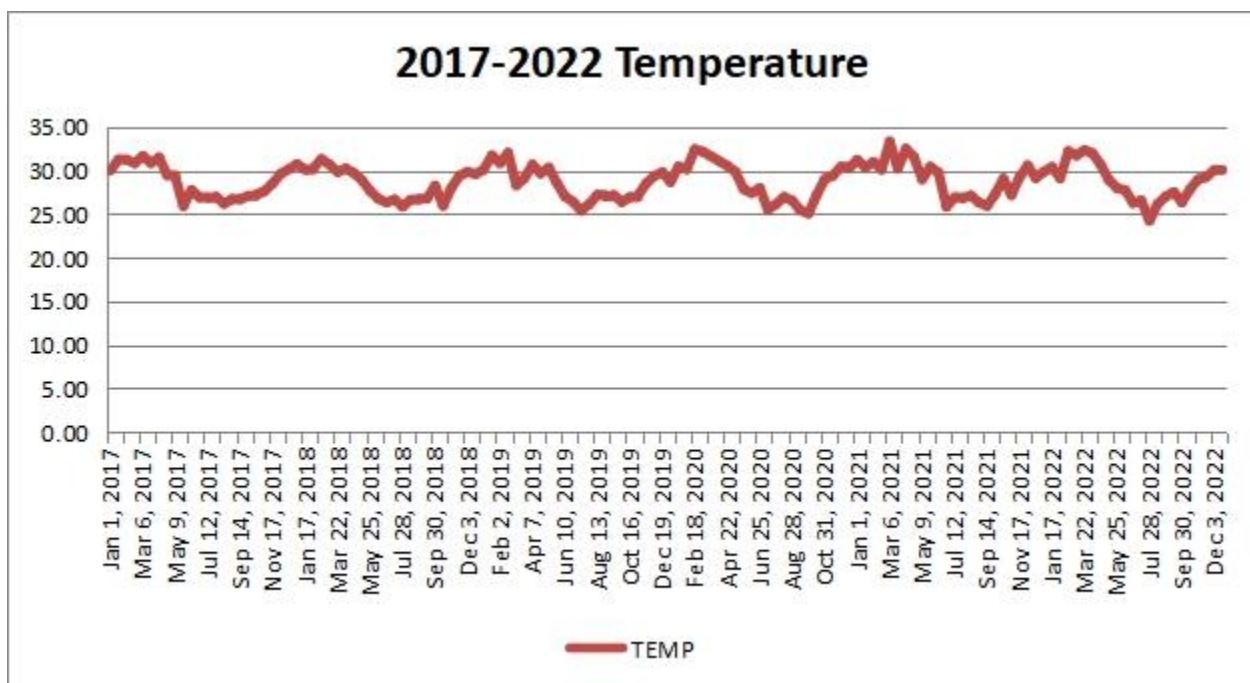


Figure 5: Temporal trend in vegetation cover from 2017-2022



For NDVI (Normalized Difference Vegetation Index), the trend plot shows moderate fluctuations across the five-year period, with values generally oscillating between approximately 3 (0.3) and 7 (0.7). This variability indicates periodic changes in vegetation greenness, which could correspond to seasonal cycles or other factors influencing plant productivity. Despite these fluctuations, the overall pattern appears relatively stable, suggesting that the ecosystem maintained consistent vegetation cover throughout the study period, with no abrupt increases or decreases in NDVI values.

In contrast, the Temperature trend shows less fluctuation in terms of extremes, with values ranging between approximately 24°C and 33°C. This relatively narrow range suggests stable temperature conditions across the years, with slight seasonal or inter-annual variations. Peaks and troughs in temperature align with expected seasonal trends, but no dramatic shifts are evident in the temperature profile, implying a stable climatic environment over the analyzed period.

Discussion

The mean temperature of 28.78°C observed during the study period suggests a predominantly warm climate within the Nigeria Forest-Savanna Ecotone, consistent with the region's transitional nature between forest and savanna ecosystems. The standard deviation of 2.01°C reflects moderate temperature fluctuations, indicating some degree of seasonal or inter-annual variability but without extreme temperature shifts. These findings align with prior studies on tropical transitional zones, where temperature variations are influenced by seasonal shifts in rainfall and vegetation cover (Ametsitsi, et al., 2020; Ike & Eludoyin, 2013; Hennenberg, et al., 2008). The observed

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temperature range (24.25°C to 33.35°C) highlights the potential for seasonal dynamics, such as dry and wet periods, to influence local microclimatic conditions.

NDVI, with a mean value of 0.49 and a standard deviation of 0.1, suggests moderate and relatively stable vegetation density throughout the study period. These values imply that the ecological conditions of the study area did not experience drastic fluctuations in vegetation productivity, which might result from consistent climatic conditions or limited anthropogenic interference. The minimum NDVI of 0.24 and maximum of 0.68 indicate variability in vegetation cover likely driven by seasonal patterns, such as wet-season vegetation growth and dry-season senescence, a pattern commonly reported in tropical ecosystems. The relatively narrow spread in both temperature and NDVI data over the five years suggests resilience in the local ecosystem, with no extreme changes in either climatic or vegetation dynamics. This stability could be attributed to the buffering capacity of mixed forest-savanna landscapes, as reported in studies examining the role of vegetation in moderating microclimate variability (Igbawua, Zhang, Chang, & Yao, 2016). The results also emphasize the importance of vegetation cover in regulating local temperatures, as areas with higher NDVI often experience reduced temperature extremes due to evapotranspiration and shading effects (Guha & Govil, 2020, Alabi & Enete, 2013).

The analysis conducted over the period from 2017 to 2022 aimed to assess the relationship between NDVI (Normalized Difference Vegetation Index) and temperature within the Nigeria Forest-Savanna Ecotone. The linear regression model indicated that NDVI explained only 0.34% of the variance in temperature, suggesting a negligible effect of vegetation cover on temperature variation. The ANOVA results further supported this, with an F-value of 0.47 and a p-value of 0.496, indicating that the effect of NDVI on temperature is not statistically significant. These

International Journal of Climate Change and Environmental Sustainability findings contrast with studies that have observed a positive correlation between vegetation cover and temperature regulation. For instance, research has shown that areas with higher NDVI often experience reduced temperature extremes due to evapotranspiration and shading effects (Guha & Govil, 2020, Alabi & Enete, 2013). Additionally, studies have highlighted the role of vegetation in moderating microclimate variability, particularly in mixed forest-savanna landscapes (Guo, et al., 2020). The lack of a significant relationship in this study may be attributed to several factors. The study period of five years might be insufficient to capture long-term trends and interactions between vegetation and temperature. Moreover, the Nigeria Forest-Savanna Ecotone is characterized by complex interactions between forest and savanna ecosystems, which may influence the dynamics of vegetation and temperature in ways not fully captured by the model. Furthermore, other climatic and environmental factors, such as precipitation patterns and soil moisture, could play a more substantial role in temperature variation within this region.

Conclusion

This study emphasizes the limited predictive relationship between vegetation greenness and temperature in this context and highlights the potential for other climatic variables to play a more significant role in temperature variation over time. The complex relationship that exist between vegetation cover and climate variables need to be further investigated as this will highlight the main factors driving vegetation changes as well regulators of climate in this region. The implications of this are not limited to food insecurity and hunger but as well will lead to inter-communal crisis over the limited shared resources. Future research may benefit from exploring additional variables to better understand the dynamics between vegetation health and climate in this ecosystem.

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